Total Printed Pages: 04

Roll No. :

CHB-101
Examination Dec - 2023
B.Tech. I/II Sem : CE, CSE, EE, EI, EC,BC, IT, IoT, AIADS
Subject: Applied Chemistry

Time :2Hrs

Max. Marks :60
Min. Marks :19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c &d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	Following two questions have four options, out of which only one is correct. Write the correct answer only. (i) Alkalinity in water cannot be due to presence of (1) OH^- only (2) OH^- and CO_3^{--} togteher (3) OH^- and HCO_3^- togteher (4) CO_3^{--} and HCO_3^- togteher (ii) Which of the following dissolved salts in water would cause the least $CaCO_3$ equivalent hardness in water sample? (1) 10 ppm of $CaSO_4$ (2) 10 ppm of $CaCO_3$ (3) 10 ppm of $MgCl_2$ (4) 10 ppm of $MgSO_4$	02	1	1
	(b)	Explain why it is not possible to remove Permanent hardness by either boiling or adding lime alone	02	3	1
	(c)	What is scale? How it is formed in boilers?	03	1	1
		OR			
		Explain Calgon conditioning giving reactions involved.	03	1	1
	(d)	Discuss Permutit (Zeolite) method of water softening giving diagram, reactions involved, merits and demerits.	05	2	1
		OR			
		100 ml of water sample required 25 ml of N/50 H_2SO_4 for neutralisation up to Phenolphthalein end point. After this, Methyl orange indicator was added to same solution and further acid required was again 25 ml. Calculate the type and extent of alkalinity of water as $CaCO_3$ equivalent.	05	3	1

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Q.2	(a)	Following two questions have four options, out of which only one is correct. Write the correct answer only. (i) Electrode potential of a metal is the tendency 1. to gain electrons 2. to lose electrons 3. either to lose or to gain electrons 4. neither to lose nor to gain electrons (ii) Which type of battery is used in Mobile phones/Laptops (1) Galvanic cell (2) Lead Accumulator battery (3) Alkaline Battery (4) Li ion	02	1	2
	(b)	Fill in the blanks: (i) A cell which converts chemical energy of fuels into electrical energy is called ..... (ii) Common cell used in wall clocks is a type of ..... battery	02	5	2
	(c)	Calculate the emf of the following cell at 25°C which is represented as $Zn_{(s)} \mid Zn^{2+}_{(0.2M)} \mid\mid Ag^{+}_{(0.002M)} \mid Ag_{(s)}$ The standard emf of the cell is 1.54 V	03	2	2
		OR			
		Electric Vehicle batteries are environment friendly over Internal Combustion Engines. Justify.	03	3	2
	(d)	Define Electrode Potential. State and derive Nernst's equation to show the dependency of Electrode Potential on concentration and temperature of solution.	05	2	2
		OR			
		Discuss construction, working and advantages of Lithium Ion Batteries.	05	6	2
Q.3	(a)	Following two questions have four options, out of which only one is correct. Write the correct answer only. (i) Food stuff containers should not be ..... from inside 1. Lead coated 2. Tinned 3. Electroplated 4. Gold coated (ii) During Galvanic corrosion, more noble metal acts as 1. Anode 2. Cathode 3. Anode as well as Cathode 4. Corroding metal	02	1	3
	(b)	Fill in the blanks: (i) Rusting of iron is higher in ..... air as compared to that in dry air. (ii) During electrochemical corrosion, the corrosion occurs at the..... part.	02	5	3

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	(c)	Rusting of iron is faster in saline water than in ordinary water.	03	2	3
		OR			
		Explain Pilling Bedworth rule.	03	2	3
	(d)	How alloying and electroplating methods are used in protection of metals from corrosion? Elaborate with examples.	05	2	3
		OR			
		What is corrosion? Discuss various factors which influence corrosion.	05	2	3
Q.4	(a)	Following two questions have four options, out of which only one is correct. Write the correct answer only. (i) Which of the following is used for making rechargeable batteries? 1. Polyaniline 2. Polyester 3. Polyacrylonitrile 4. Polypyrrole (ii) Basic principle of Optical Fibers is 1. Optical rotation 2. Scattering of light 3. Total Internal absorption 4. Total internal Reflection	02	1	4
	(b)	What are nanomaterials? Write their significance.	02	5	4
	(c)	What are Liquid Crystal polymers? How do they work? Write their applications.	03	4	4
		OR			
		What are polymers? Give classification of polymers giving suitable examples.	03	2	4
	(d)	What are solar cells? How do they work? Write the applications of solar cells.	05	2	4
		OR			
		Write a short note on following: 1. Applications of nanomaterials 2. Conducting polymers	05	2	4
Q.5	(a)	Following two questions have four options, out of which only one is correct. Write the correct answer only. (i) Technique of Gas Chromatography is dependent on difference in ..... of two or more solutes 1. Boiling Point 2. Rate of Adsorption 3. Magnetic moment 4. Solubility (ii) IR spectroscopy is used in ..... of organic compounds 1. Quantitative analysis 2. Separation 3. Identification and characterization 4. Molecular weight determination	02	1	5
	(b)	Fill in the blanks: 1. Hertz is the unit of ..... 2. pH scale ranges from ..... to		5	5

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(c)	State and explain Beer Lambert's law giving mathematical expression.	03	2	5
	OR			
	What is pH? Calculate pH value of 0.1 M solution of NaOH.	03	2	5
(d)	Define conductivity. Discuss conductometric titration of strong acid v/s strong base giving the experimental set up and the curve (graph) obtained.	05	2	5
	OR			
	Define following terms: 1. R_F value 2. Electrolyte 3. Precision and Accuracy 4. Elution 5. Absorbance	05	2	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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